Monitoring the Fate of Zn in Cu/ZnO/ZrO_2 during CO_2 to Methanol Synthesis at High Conversions in a Novel 3D-printed Steel Reactor

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Motivation

The Power-to-Liquid (PtL) concept represents a promising approach for storing electrical energy from renewable sources by converting green H_2 and CO_2 to liquid fuels and chemicals, such as methanol (MeOH). However, CO_2-based MeOH synthesis is more challenging compared to the conventional process from CO-rich syngas. The higher water content in the product stream accelerates deactivation of the conventional Cu/ZnO/Al_2O_3 (CZA) catalyst and the thermodynamic equilibrium limits the methanol yield. The *in situ* condensation and separation of MeOH (and H_2O) by using two temperature zones in the reactor can effectively shift the equilibrium to the product side.[1] However, condensation inside the catalyst bed is a high risk for catalyst deactivation. The structure of Cu-Zn based catalyst system is further known to be highly dynamic as it changes during activation, reaction and deactivation. However, *operando* spectroscopic investigations at high conversion are lacking due to limitations of the used sample amount, dilution and high pressure. Recently, restructuring induced by the presence of alcohol, *e.g.* methanol, was proposed for a Cu/ZnO/Al_2O_3 catalyst,[2] further highlighting the need for investigations with significant product amounts, typical for the end of a fixed bed.

Experimental part/methodology[3]

To address this need, we have developed a multifunctional reactor fabricated by additive manufacturing, enabling high-pressure (up to 90 bar), *operando* X-ray absorption spectroscopy (XAS) studies thanks to X-ray transparent windows, and two

compartments for a dual catalyst bed concept with independent temperature control (Fig. 1). The first larger catalyst bed ensures sufficient liquid products to investigate the structure during high conversion and significant MeOH and water concentrations. Cooling liquid can be introduced around the downstream catalyst bed and the X-ray windows to induce condensation and monitor the effect on the catalyst structure using *operando* X-ray absorption spectroscopy. For this investigation, a Cu/ZnO/ZrO_2 catalyst with similar composition to the commercial CZA catalyst and high activity in CO_2 hydrogenation was used.

Results and discussion[3]

Operando XAS in the newly developed reactor during CO_2-based methanol synthesis at 90 bar, as well as during induced condensation of the reaction products, revealed pronounced dynamics in catalyst structure and performance: While the state of Cu remained mostly unchanged, Zn showed highly dynamic changes. We observed partial dispersion of ZnO during high conversion. Particularly, condensation has unfolded to be critical, as in this case irreversible $ZnCO_3$ formation has been observed (Fig. 1). This study shows the importance to investigate catalysts both at low and high conversion, which becomes possible with the dual compartment strategy and provides insights into the active state of a MeOH catalyst under high conversion conditions. Finally, it uncovered deactivation pathways caused by condensation of the reaction products.

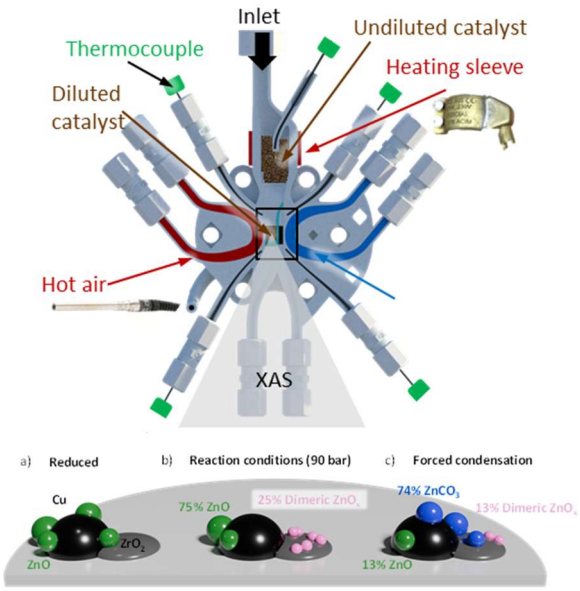

Figure 1: Schematic illustration of the multifunctional reactor (top) and present Zn species after a) reduction, b) reaction conditions at 90 bar and c) forced condensation.

References

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